

Yong Chang Go — Resume



Profile

Software engineer with 3+ years of experience in semiconductor manufacturing IT, building automation tools, desktop applications, and web systems that support fab operations. Currently a Software Engineer at VSMC focused on engineering data analysis and defect management systems. Previously built MES-integrated tools at UMC handling real-time lot tracking, rack inventory, and legacy system migrations.

Employment History

Software Engineer, Data Analysis Platform — VisionPower Semiconductor Manufacturing Company (VSMC)

June 2025 — Present

- Build and maintain a **KLA Map-to-Map Data Watchdog** in Python — an automated SPC pipeline that monitors active lots in real-time, calculates DC values via formula-driven SQL against KLA inspection data, and submits results to MES with Oracle audit logging and automated retry on failure — processing ~10 lots/hour autonomously, saving an estimated 1.5–2 engineering hours per hour of operation that would otherwise be spent on manual data retrieval and calculation
- Design and develop systems for fab engineering data analysis, defect management, and automated watchdog jobs
- Gather requirements directly from engineers and lead end-to-end system delivery — from design through deployment
- Evaluate and introduce new tooling to reduce manual data processing and improve operational efficiency

Software Developer, Manufacturing Execution Systems — United Microelectronics Corporation (UMC), Singapore Branch

February 2023 — June 2025

- Migrated and enhanced the **eRack Management System** from a legacy ASP.NET web application to a C#/.NET Windows Forms desktop application with Oracle SQL and multi-threading — enabling real-time MES sync via transaction calling, reducing FOUP check-in/check-out time by ~50% compared to the manual process, and improving overall usability
- Built an **Automated WAT Rework and Backup System** in C# integrated with Aegis and ADO.NET, automating lot verification, hold code management, and parameter modifications — processing up to 50 lots/hour (vs ~30 minutes per lot manually), reducing issue-checking time by 80% through

automated alarm email notifications, and eliminating manual errors via automated hold comments and error holding

- Migrated the **Reticle Management System** from VB to C#/ASP.NET Web Forms, implementing AJAX partial updates and session management — improving overall application performance, usability, and boosting daily operation efficiency by ~30%
- Maintained and enhanced existing MES applications, collaborating with production staff to diagnose issues and deploy fixes with minimal downtime

IT Support Intern — Vis-Dynamics Research Sdn Bhd, Malacca

March 2021 — July 2021

- Provided vendor IT support across networking and server infrastructure; implemented security hardening measures
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Technical Skills

Languages: C#, Python, SQL, HTML/CSS/JavaScript, VB.NET, SAS

Frameworks: .NET Framework, ASP.NET Web Forms, Windows Forms, ADO.NET

Tools & Platforms: Git, Gitea, Microsoft Azure DevOps, DBeaver, Oracle SQL Developer

Domain Expertise: Semiconductor Manufacturing, MES Integration, SPC Data Pipelines, Process Automation

Education

Bachelor of Information Technology (Honours) — Artificial Intelligence

Multimedia University (Malacca Campus), Malacca

June 2018 — September 2022 | CGPA: 3.36